

GLT 101



Heating: Use a clamp or tongs to hold hot objects. Do not touch hot objects with your bare hands.



Sharp Object: Scissors, scalpels, pins, and knives are sharp. They can cut or puncture your skin. Always direct sharp edges and points away from yourself and others. Use sharp instruments only as directed.



Electric Shock: Avoid the possibility of electric shock. Never use electrical equipment around water, or when the equipment or your hands are wet. Be sure cords are untangled and cannot trip anyone. Disconnect equipment when it is not in use.



Poison: Do not let any poisonous chemical get on your skin, and do not inhale its vapor. Wash your hands when you are finished with the activity.



Physical Safety: This activity involves physical activity. Use caution to avoid injuring yourself or others. Follow instructions from your teacher. Alert your teacher if there is any reason that you should not participate in the activity.



Animal Safety: Treat live animals with care to avoid injuring the animals or yourself. Working with animal parts or preserved animals may also require caution. Wash your hands when you are finished with the activity.



Flames: Tie back loose hair and clothing, and put on safety goggles before working with fire. Follow instructions from your teacher about lighting and extinguishing flames.



No Flames: Flammable materials may be present. Make sure there are no flames, sparks, or exposed sources of heat present.



Fumes: Poisonous or unpleasant vapors may be produced. Work in a well-ventilated area. Avoid inhaling a vapor directly. Test an odor only when directed to do so by your teacher, using a wafting motion to direct the vapor toward your nose.



Disposal: Chemicals and other materials used in the activity must be disposed of safely.



Hand Washing: Wash your hands thoroughly when finished with the activity. Use antibacterial soap and warm water. Lather both sides of your hands and between your fingers. Rinse well.



Biohazard: This symbol indicates that there is some form of biological hazard. Use with extreme care or avoid if necessary.



Radiation: This symbol indicates that there is some sort of radiation near. Proceed with caution or avoid if necessary.

2.2 ACID BURNS

2.2.1 Acetic acid, glacial (poisonous corrosive);

Acute Effects: Ingestion may be fatal! Symptoms include sore throat, vomiting and diarrhea. Ingestion of 1.0 milliliter (ml) may cause esophageal perforation. Vapor inhalation damages the respiratory tract lining and causes breathing difficulty. Concentrations are inadequately detected by odor or irritation. Contact may cause serious skin damage. Concentrated solutions making eye contact may cause blindness. Vapors irritate eyes. **Chronic:** May cause skin darkening, dental erosion and chronic upper respiratory tract inflammation.



First Aid: Get medical help! For inhalation, move victim to fresh air and give respiratory aid if needed. For contact, wash affected area with large amounts of water (at least 15-20 minutes). If ingested, do not induce vomiting. Give large quantities of milk or water.

Flammable: Combustible.

Cleanup: Ventilate and evacuate the area. Wear protective gear. Shut off ignition sources. Use water spray to disperse vapors. Dilute to nonflammable mixtures with water. Absorb with an inert material. Place in a closed container for disposal. Use non sparking tools. Avoid flushing to sewer.



Acetone (flammable): In chromatography solvent, spectrophotometer blanks and pigment extracts. Toxic effects may be enhanced if alcoholic beverages are used.

Acute: Inhalation irritates nose, throat and mucous membranes. Inhalation or ingestion may cause dizziness, dullness and headache. May act as a narcotic. Contact defats skin, causes redness, pain, drying and cracking. Eye contact is irritating. **Chronic:** Severe irritation or dermatitis.

First Aid: If inhaled, move to fresh air and give respiratory aid if needed. If ingested, and victim is conscious, give 2 glasses of water and induce vomiting. For contact, wash area for at least 15 minutes with large amounts of water. Remove contaminated shoes and clothing. Get medical aid.

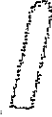
Flammable: Vapor/air mixtures above the flash point [10°F] may be explosive. Use water to flush spills and to dilute to non-flammable concentrations. Incompatible with oxidizing materials, chloroform, alkalies, acids and potassium t-butoxide.

Clean-up: Wear protective gear for toxic vapors (goggles, rubber gloves and boots, respirator, apron). Remove ignition sources and isolate the spill. Recover liquid for reclamation, or absorb with an inert material and hold for disposal. Do not flush to sewer. Do not wear contact lenses.



AMMONIUM HYDROXIDE (POISONOUS CORROSIVE): Exceptional health and contact hazard! May be fatal if swallowed, causes burns and vapor is extremely irritating.

Acute Health Effects: Inhalation may severely irritate or burn the respiratory tract, cause pulmonary edema and inflammation of the lungs. Contact may cause severe irritation and burns. Eye contact may cause permanent corneal damage and blindness. Ingestion may severely burn the mouth and stomach and may be fatal.

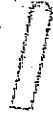


First Aid: Call a physician! If ingested, **do not induce vomiting!** If conscious, give large amounts of water followed by diluted vinegar, fruit juice or egg whites beaten with water. If inhaled, move victim to fresh air and provide respiratory aid if needed. For contact, flush area with large amounts of water for at least 15 minutes. Remove contaminated clothing and shoes and wash before reuse.



Flammable: Flammable vapors may form an explosive mixture with air. Produces toxic ammonia and hydrogen gases if ignited. Incompatible with acids, alkali metals, strong oxidizing agents, bromine, chlorine, aluminum, copper, brass, bronze, mercury and dimethyl sulfate.

Clean-up: Wear full protective clothing and a respirator. Stop leak if it can be done safely. Ventilate area. Neutralize with dilute HCL, vinegar or commercial caustic neutralizer and flush to sewer.



Benedict's (corrosive): Contains sodium citrate, dihydrate, sodium carbonate and cupric sulfate. Persons with Wilson's disease should avoid contact.



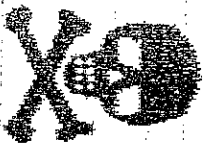
Acute Effects: Inhalation irritates nasal membranes and upper respiratory tract, causes coughing, nausea, vomiting, sore throat and pulmonary edema. Skin contact may cause itching, redness and dermatitis. Eye contact may result in blurred vision, burns, conjunctival edema and corneal opacification. Ingestion may cause nausea, vomiting, headache, ataxia, dizziness, abdominal pain, bloody diarrhea, esophageal perforation and salivation. Chronic Effects: Irritation of the upper respiratory tract, dermatitis and conjunctivitis. Ingestion impairs calcium and iron absorption. May be fatal in persons with Wilson's disease.

First Aid: If inhaled, move victim to fresh air and give respiratory aid if needed. For contact, wash area with large amounts of water (at least 15-20 minutes). If ingested give water or milk and allow vomiting to occur (do not induce vomiting). Examine mouth and throat for corrosive injury, irrigate affected areas with a 1% solution of acetic acid until alkali is neutralized. Avoid gastric lavage or emesis. Maintain airway and treat for shock. If vomiting occurs, keep head below hips to help prevent aspiration. Get medical aid!

Flammable: Not flammable. It is a reactivity hazard (check MSDS).

Clean-up: Treat aqueous solution spills as a corrosive alkali (base).

Hold all waste in a clean container for proper disposal.



Chloroform (poison, possible carcinogen): In chromatography solvent. DNA isolation labs. Wash thoroughly after handling. Persons with disorders of the skin, eyes, liver, kidney or respiratory system may be more susceptible to the effects of this substance.

Acute Health Effects: May be fatal if ingested, inhaled or absorbed! Inhalation causes respiratory tract irritation, headache, drowsiness, vomiting, dizziness and unconsciousness. Ingestion burns the mouth and throat, causes chest pain, vomiting and unconsciousness. Skin contact is irritating. Eye contact may irritate the conjunctiva.

Chronic Health Effects: May cause death from heart arrhythmias and kidney or liver damage. May also cause dermatitis and malignant tumors.

First Aid: If inhaled, move victim to fresh air and give respiratory aid. If ingested, and victim is conscious, give 2 glasses of milk or water and induce vomiting. For contact, rinse area with large amounts of water for at least 15 minutes. Remove contaminated clothing. Get medical aid.

Flammable: Slight fire hazard at high temperatures. Thermal decomposition may release toxic gases (carbon monoxide, hydrogen chloride, chlorine, phosgene). Incompatible with strong caustics and chemically active metals (aluminum, magnesium, sodium, potassium), acetone, fluorine, methanol, sodium methoxide, dinitrogen tetroxide, tert-butoxide and triisopropylphosphine.

Clean-up: Wear protective gear suitable for toxic fluids or vapors (safety goggles, respirator, rubber gloves, boots and apron). Ventilate the area and isolate the spill. Remove ignition sources. Recover the liquid, or absorb with an inert material and collect as hazardous waste. Do not flush to sewer. Do not wear contact lenses.



Ether (diethyl ether) (poison, flammable): In spectrophotometer blanks and pigment extracts.

Acute Health Effects: May be fatal if inhaled or ingested! Inhalation may also cause headache, nausea, vomiting, dizziness, narcosis, suffocation, low blood pressure and central nervous system depression. Readily absorbed on contact, may irritate eyes. Ingestion may cause nausea, vomiting and central nervous system depression.

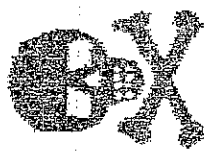
Chronic Health Effects: Skin sensitization.

First Aid: If inhaled, move to fresh air and give respiratory aid if needed. If ingested, give milk or water and induce vomiting. For contact, wash skin or eyes for at least 15 minutes with large amounts of water. Get medical attention.



Flammable: Flash point is -49°F. Extremely flammable and highly volatile. May form explosive peroxides if heated or in an anhydrous state. Use alcohol foam, dry chemical or carbon dioxide to extinguish (water may be ineffective). Vapors can travel along surfaces to distant ignition sources and flash back. Containers exposed to heat may explode. Contact with oxidizers may cause fire. Produces toxic carbon monoxide and carbon dioxide gases. Incompatible with strong bases, strong oxidizing agents, aluminum, and combustible materials.

Clean-up: Wear protective gear suitable for toxic fluids or vapors. Shut off ignition sources and isolate the spill. Use water spray to reduce vapors. Absorb with an inert material and place in a container for later disposal. Flush area with water.



Ethanol (poisonous, flammable solvent):

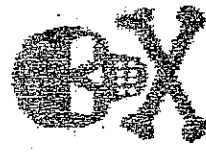
Acute Effects: Ingestion may be fatal. May be absorbed through the skin. Vapor and mist irritates the eyes, mucous membranes and upper respiratory tract. May cause eye damage, nausea, dizziness, headache and central nervous system depression. Chronic: Liver, heart and kidney damage.

First Aid: For contact, flush with large amounts of water for at least 15 minutes. Remove contaminated clothing and shoes, and wash before reusing. If inhaled, move to fresh air and give respiratory aid if needed. For ingestion, give large amounts of water and call a physician.



Flammable: Flash point 48°F. Use dry chemical, carbon dioxide, alcohol or polymer foam. Vapor may travel to distant ignition source and flash back.

Clean-up: Shut off ignition sources and evacuate the lab. Wear rubber boots and rubber gloves. Absorb with an inert material and place in a closed container for disposal. Use non-sparking tools. Ventilate area and wash spill site after material pickup is complete.



Hydrochloric acid (poisonous corrosive): Liquid and mist may severely burn all tissues. May be fatal if swallowed. Avoid contact, wash thoroughly after handling. Persons with pre-existing skin or eye disease may be more susceptible to the effects of this substance.

Acute Health Effects: Corrosive! Vapor inhalation may cause coughing, choking and inflammation of the nose, throat and upper respiratory tract. Vapors may cause lung damage. Ingestion causes immediate pain and burns the mouth, throat, esophagus and gastrointestinal tract. May cause nausea, vomiting and diarrhea.



Contact may cause redness, pain and severe burns. Concentrated HCL causes deep ulcers and skin discoloration. Vapors and splashes are irritating and may severely burn and damage eyes. Chronic Health Effects: Vapor exposure may

erode teeth. Because of its corrosive properties, chronic exposures are rare.

First Aid: If **inhaled**, move victim to fresh air and give respiratory aid if needed. If **ingested**, do **not induce vomiting!** If victim is conscious, give large quantities of milk or water. For **skin contact**, wash at least 15 minutes with large amounts of water. Remove contaminated shoes and clothing and wash before reuse. For **eye contact**, flush with water at least 15 minutes or until no trace of chemical remains. Get medical help!

Flammable: Reacts with metals to release flammable hydrogen gas and when heated, releases toxic hydrogen chloride. Use water spray to extinguish. Highly reactive with strong bases, metals, metal oxides, hydroxides, amines, carbonate and other alkaline materials. Incompatible with cyanides, sulfides, sulfites and formaldehyde.

Clean-up: Wear protective gear suitable for toxic or corrosive fluids or vapors (goggles, respirator, rubber gloves and boots, apron etc.). Isolate the spill. For small spills, neutralize with alkaline material (soda ash, lime etc) and flush to the sewer with excess water. For large spills, neutralize with alkaline material and pick-up with absorbent material. Provide forced ventilation to dissipate fumes and dispose in an approved waste facility or sewer the neutralized slurry with excess water if local ordinances allow.



Sodium hydroxide (poisonous corrosive): Strongly exothermic when solid contacts water. Persons with disorders of the respiratory tract, skin or eyes may be more susceptible.

Acute Effects: **Strong corrosive!** Ingestion may burn mouth, throat and stomach, cause scarring and death. Skin contact may burn and scar. Contact with eyes is irritating and may cause blindness. **Inhalation** damage varies from mild irritation to serious upper respiratory tract damage depending upon exposure. Severe pneumonitis may occur. **Chronic Health**

Effects: Prolonged contact with dilute solutions or dust destroys tissue.

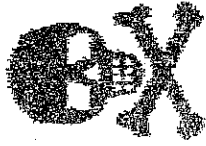


First Aid: If **inhaled**, move victim to fresh air and give respiratory aid if needed. If **ingested**, do **not induce vomiting!** If victim is conscious, give 2-4 glasses of milk or water and get medical help. For **skin contact**, brush dry chemical off to limit thermal reaction with water and skin moisture. Remove contaminated clothing and flush for at least 15 minutes with large amounts of water. For **eye contact**, flush with water until no trace of chemical remains (at least 15 minutes).

Flammable: Reacts with aluminum, tin and zinc to form flammable hydrogen gas. Contact with water generates large amounts of heat. Incompatible with water, acids, flammable liquids and organic halogen compounds which may cause fire or explosion. Contact with nitro compounds causes shock-sensitive salts to form.

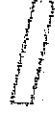
Clean-up: Wear protective gear. Do not wear contact lenses. Sweep, scoop or pick up dry material; avoid raising dust. Place into a closed, metal, container and send to an approved waste disposal facility. Do not flush to the sewer avoid contact with water. Neutralize chemical traces

on dry spill site with dilute acid and flush to the sewer with excess water. Absorb aqueous spills with caustic neutralizer and flush neutralized slurry to the sewer with excess water.



Sulfuric acid (poisonous corrosive): Liquid and mist may severely burn tissues. May be fatal if ingested. Persons with skin or eye disease may be more susceptible to the effects of this substance.

Acute Health Effects: Corrosive! Vapor inhalation may severely irritate respiratory tract. Ingestion may be fatal. Causes immediate pain and burns the mouth, throat, esophagus and gastrointestinal tract. May cause nausea and vomiting and adversely affect kidney function. Contact may cause and severe burns to eyes and skin. **Chronic** Health Effects: Vapor exposure may cause lung damage.



First Aid: If inhaled, move victim to fresh air and give respiratory aid if needed. If ingested, do not induce vomiting! If victim is conscious, give large quantities of milk or water. For skin contact, wash at least 15 minutes with large amounts of water. Remove contaminated shoes and clothing and wash before reuse. For eye contact, flush with water at least 15 minutes or until no trace of chemical remains. Get medical help!!

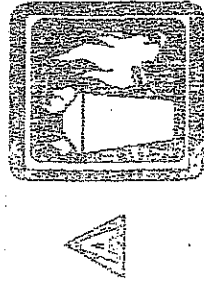


Flammable: Reacts with metals to release flammable hydrogen gas. A violent exothermic reaction occurs if mixed with water. Do not get water inside containers. Use dry chemical or carbon dioxide to extinguish fires. Incompatible with water, strong bases, metals, organic materials, strong reducing agents, combustible material and strong oxidizing agents.

Clean-up: Wear protective gear suitable for toxic or corrosive fluids or vapors (goggles, respirator, rubber gloves and boots, apron etc.). Isolate the spill. For small spills, neutralize with alkaline material (soda ash, lime etc). Use clean shovel to place material into a clean, dry container and cover. Do not use water.

2.3 Classification of Fires

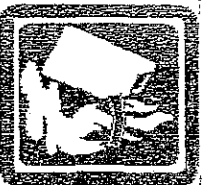
Most fires that occur will fall into one or more of the following categories:



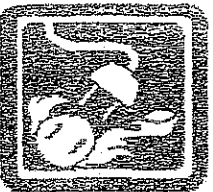
- Class A: Fires involving ordinary combustible materials, such as paper, wood, and textile fibers, where a cooling, blanketing, or wetting extinguishing agent is needed.



- Class B: Fires involving flammable liquids such as gasoline, thinners, oil-based paints and greases. Extinguishers for this type of fire include carbon dioxide, dry chemical* and halogenated agent types.



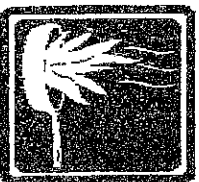
- Class C: Fires involving energized electrical equipment, where a non-conducting gaseous clean agent or smothering agent is needed. The most common type of extinguisher for this class is a carbon dioxide extinguisher.



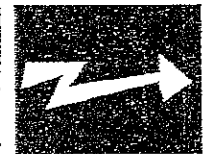
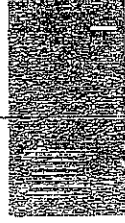
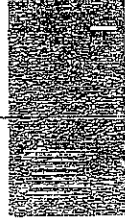
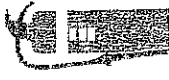
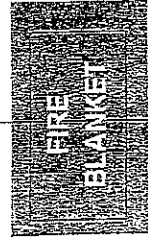
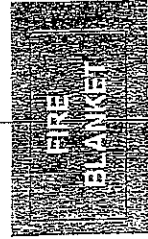
- Class D: Fires involving combustible metals such as magnesium, sodium, potassium, titanium, and aluminum. Special dry powder extinguishing agents are required for this class of fire, and must be tailored to the specific hazardous metal.

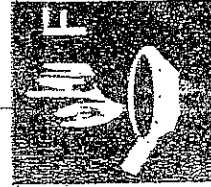


- Class K: Fires involving commercial cooking appliances with vegetable oils, animal oils, or fats at high temperatures. A wet potassium acetate, low pH-based agent is used for this class of fire.



Please take care not to confuse dry chemical extinguishers with dry powder extinguishers. Dry powder extinguishant will cake and form an exterior crust when heated by the fire. The crust formed will exclude air from the chemical reaction, resulting in extinguishment of the fire, and dissipating heat. A regular dry chemical extinguisher will simply blanket the burning material, interrupting the chemical reaction at the fuel's surface.

Old Colour BS 5406	New Colour BS EN3	A	B			
						
						
						
						
						
				General Note - May be used in conjunction with other extinguishing agents / or fire extinguishing techniques.		





SPECIALIST HOT COOKING OIL, TIRES ONLY

Specifically for dealing with high temperature (360°C+) cooking oils used in large industrial size catering kitchens, restaurants and takeaway establishments with deep fat frying facilities.

